SEPTEMBER 28/9/24

I HAVE USED UNET IMAGE SEGMENTATION MODELS WITH RESNET,

THE MAIN PROBLEM with using convolutional neural network is that it is not good at getting the global context of the images.

If were doing just the skin image lesion classification where the boundary is plain (i.e. it has not many artifacts), convolutional networks are good at these work, which is capturing local attention

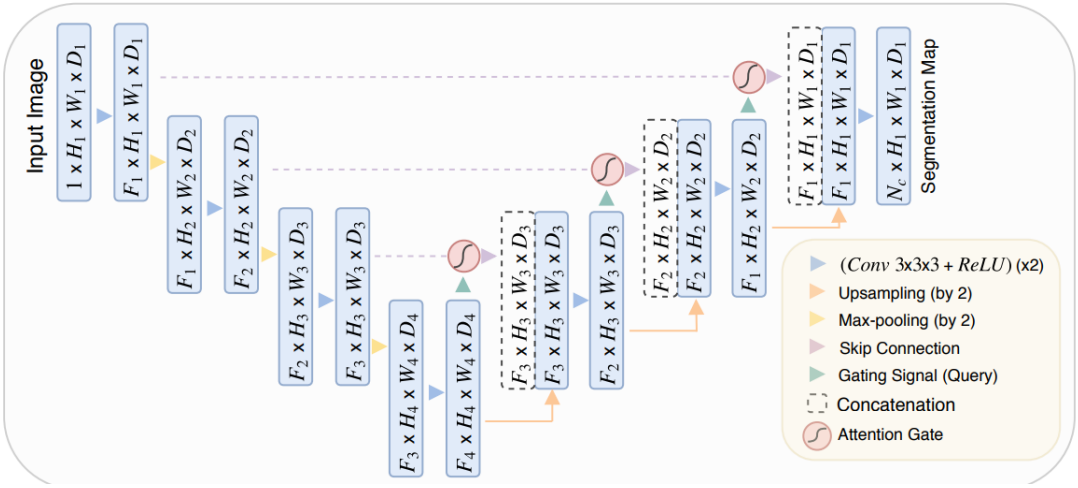
So, I looked up some papers and some internet and CHATGPT, it seems attention mechanisms are really good at getting the whole semantic information of the given image

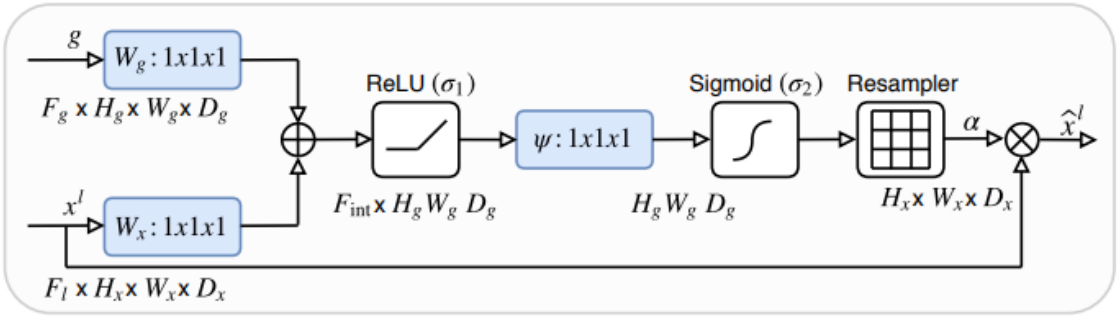
They are good because of the very humane like approach for image segmentation

We can also play with parameters which would help us in telling the network what is important and what is not important. In this way we can really make the network ignore the artifacts in the oral part keeping the context of the image in hand

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Using UNET with attention, will be using soft attention on the relevant image using this architecture





Orignally used in the paper where to look for pancreas , the reason i choose this paper is because of the similar conditions in the dataset(varying shapes)

Used focal loss as the main learning loss:

Batch size= 10

Learning rate = 0.001

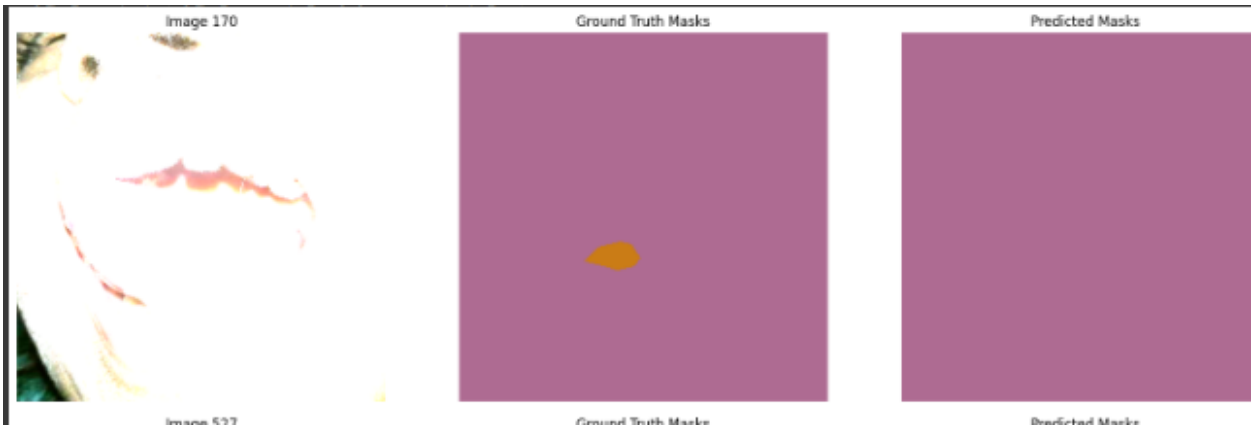
This is the result i got for some test cases:



The colors here are vague due to some clipping error(negative values in the class label)



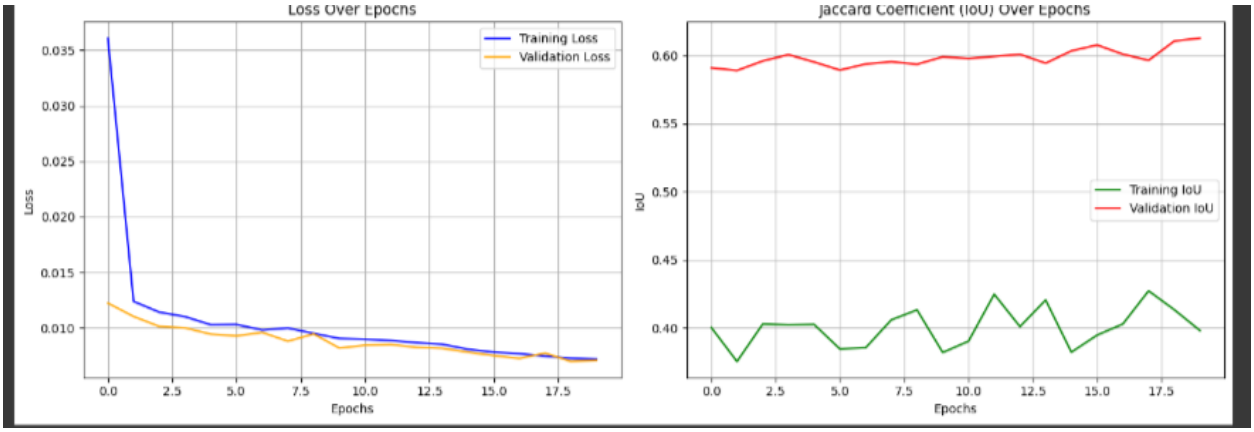
These are again some other results



Some cases where the model hasn't been working great

These are the learning curves:(Validation and Training Focal loss and for both IOU SCORE)

Made sure there is no data leakage as discussed with buddhdev sir

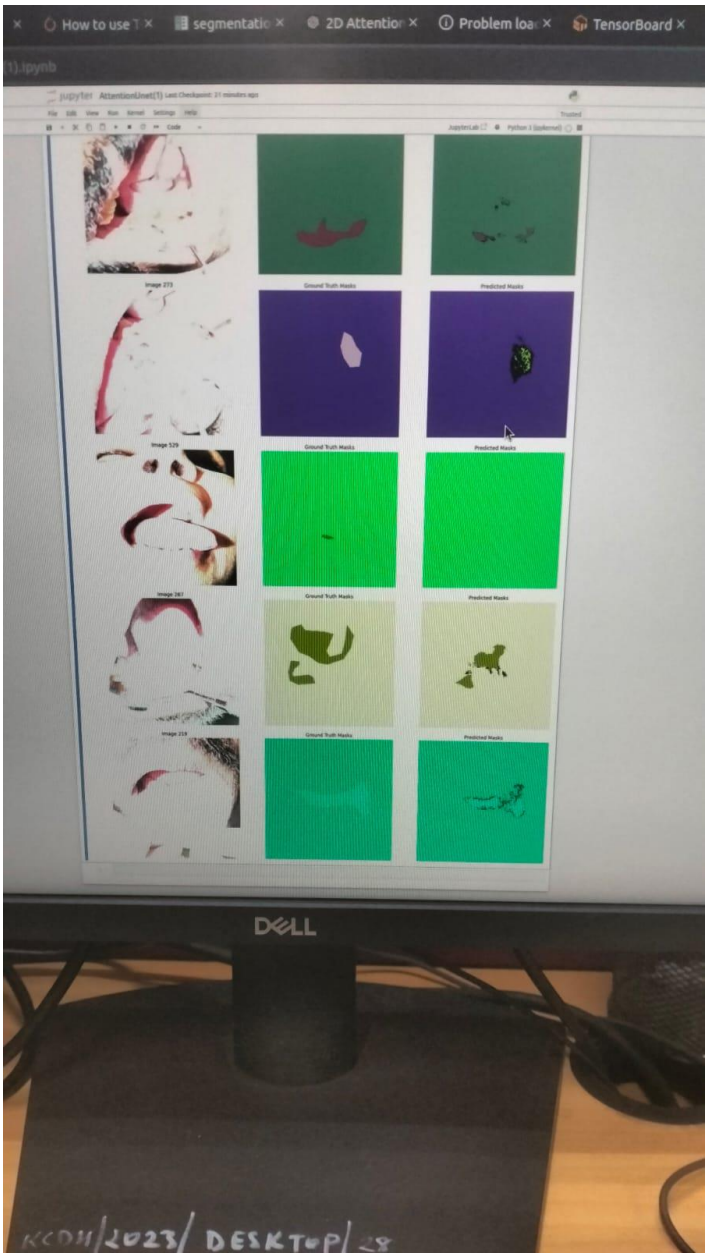


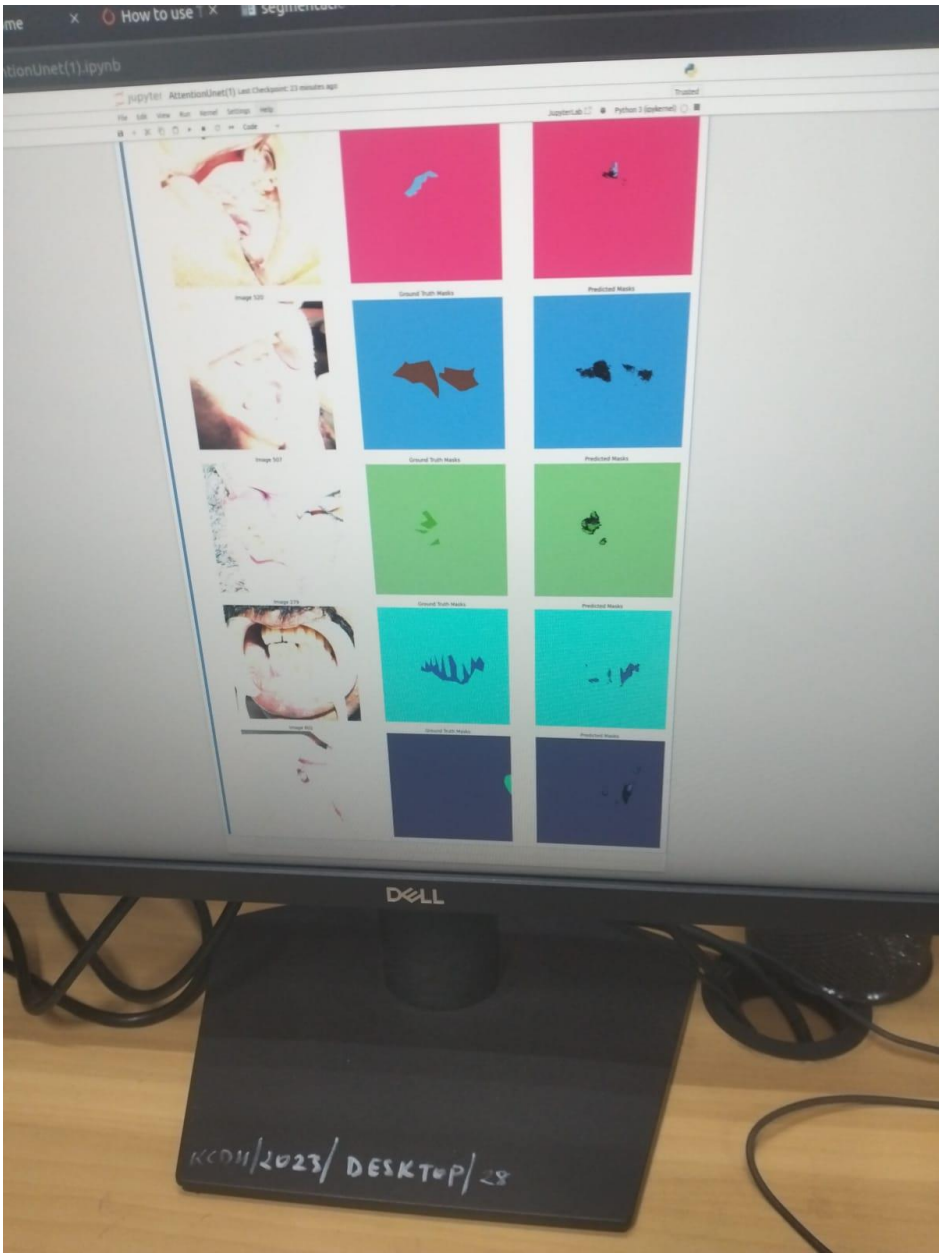
Since there is a class imbalance, i used focal loss

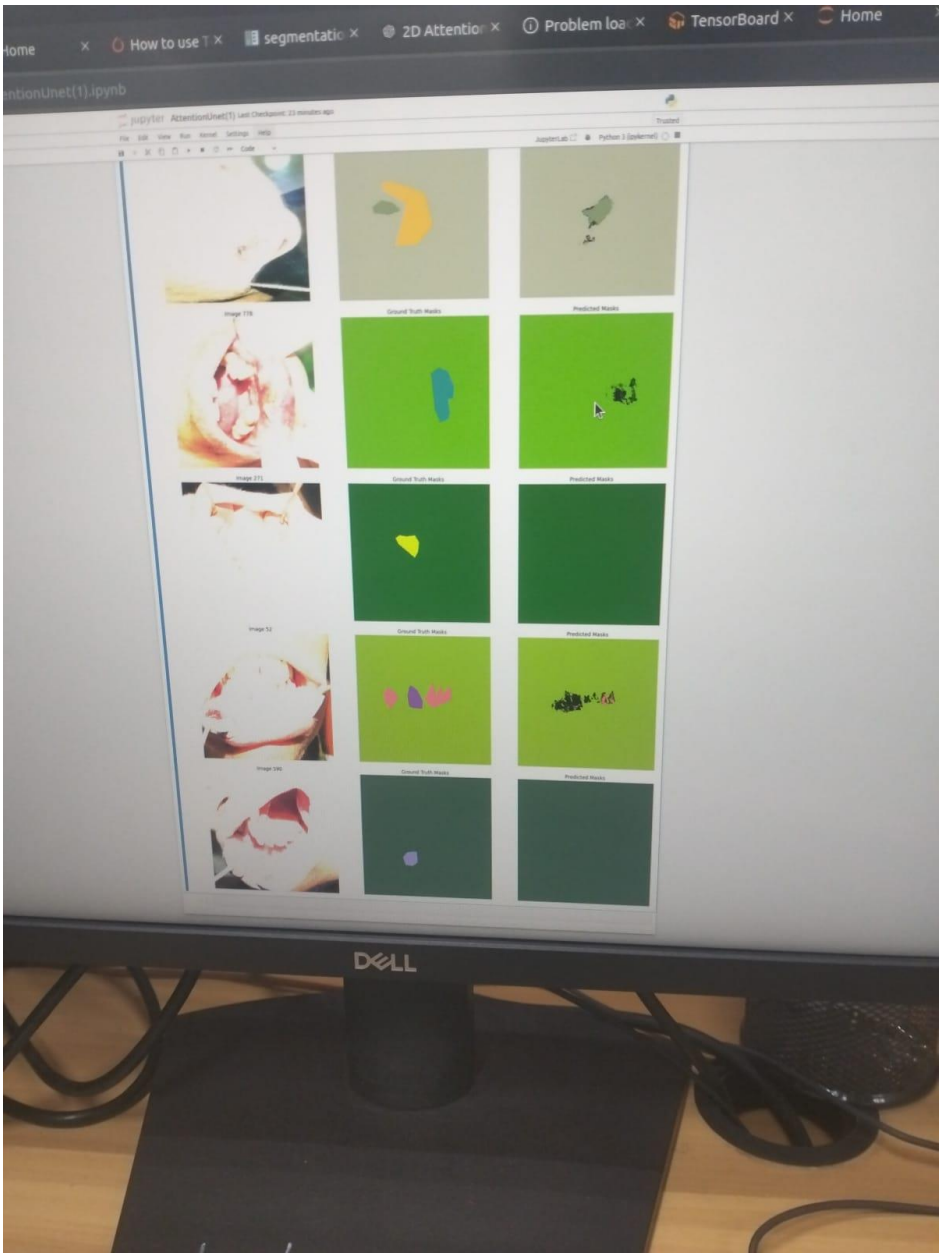
$$\mathrm{FL}(p_t) = -(1 - p_t)^{\gamma} \log(p_t)$$

As seen this is a loss which emphasizes more on the not so well classified examples and punishes the model for doing so

For evaluation metric I used IOU/Jaccard.


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